

Introduction and Data

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```
library(survival)
library(knitr)
opts_chunk$set(comment=NA, tidy=FALSE, highlight=FALSE, echo=TRUE,
               fig.with=7, fig.height=5.5, fig.path="figures/",
               out.width="\\textwidth", out.height="!", device="pdf",
               cache=FALSE, background="#FFFFFF",
               warning=FALSE)
options(width=60)
options(show.significant.stars = FALSE) # display statistical intelligence
table2 <- function(...) table(..., useNA="ifany")
```

1 Introduction

Well, I generally don't generate ideas of my own. Someone has to come to me with a problem. And, apparently, I'm pretty good at coming up with solutions or ideas for solutions. Identifying problems is what is important – solutions just follow. Nathan Mantel

History

- 1953–71 Dairy farm in SE Minn
- 1971–75 St Olaf College, Mathematics
- 1975–76 Dairy
- 1976–79 Computer programmer, Mayo Clinic
- 1979–83 PhD Stanford

- 1983–85 University of Rochester, NY
- 1985–26 Statistician, Mayo Clinic

Survival Analysis

- Taylor: all possible Cox regressions
- Moss and Bigger: should ejection fraction be linear or categorical?
- MDPIT: Can we make use of extra information?
- 1984-85: early genesis of the package
- Statlib (1987?), S-Plus, R
- Pattern: A medical problem required an analysis that I didn't have

Why am I here?

I have analyses that I have done based on somewhat ad hoc theory, where I am comfortable that I have done good science but would appreciate confirmation of the underlying method. I have other, pending analyses where things are still foggy. I bring real data (which you can use) and real problems, and you all can supply the answers.

A second issue will be coding. I have made several choices wrt multistate models in the survival package, and in the new mshmm package. Are they sensible, useful, understandable, ...?

More importantly, this will give us a chance to talk about issues that arise in real data, which I hope will be useful.

2 Semi-theory

The three most popular models in statistics are

- Linear: $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$
- GLM: $E(y) = g(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots)$
- Cox: $\lambda(t) = g(\beta_0(t) + \beta_1 x_1 + \beta_2 x_2 + \dots)$

The reason why is simplicity. If 'x1= apoeapos', then β_1 is *THE* effect of APOE, independent of any other variables in the model. Statisticians like this. Investigators really like this (a simple story). Regulators demand it — a single yes/no p-value.

- Generalized additive models will replace one of the βx terms with $s(x)$, but retain the separability.
- The function g is chosen to make the model code never fail, which means never predicting an impossible value.
 - binomial: logistic, probit, cloglog
 - Cox, poisson: exp
- This makes the models less interpretable.

Science is not always this simple, and multi-state models break the rules. That is, a description of the final fit can not be boiled down to one number.

3 Examples

Many of these will be further explained in the draft book (Atkinson, Crowson and Therneau) and its examples. These are part of the materials for the course. Some data sets are in the survival package, some in the mshmm package (github.com/therneau/mshmm).

3.1 Crohn's disease

This is discussed in the Therneau and Grambsch (2000), before I understood multistate models very well. The data has been made available in the mshmm package. I would be very interested in a more thoughtful analysis.

Crohn's disease is a chronic idiopathic disease of the gut characterized by transmural inflammation and ulceration. Management may require immunosuppressive drugs and surgical interventions. This data follows an inception cohort of 174 residents from 1970 through 1993, with a median follow-up of 10 years. The waxing and waning nature of the disease makes it challenging to analyse.

- 1488 observations on 174 subjects
- States of remission, mild, treatment responsive, treatment dependent, treatment resistant, surgery, death

3.2 Fatty liver disease

Data from a cohort of subjects with non-alcoholic fatty liver disease (NAFLD) and a set of matched controls. Nonalcoholic fatty liver disease (NAFLD) was renamed metabolic dysfunction-associated steatotic liver disease (MASLD) in 1993; the new name is intended

to better reflect the disease's underlying causes, identify subgroups of patients, and avoid stigmatizing words.

The study constructed a population cohort of all MASLD subjects from 1997 to 2014 along with 4 potential controls for each case, matched on age, sex, and calendar year. The primary analysis looked at the incidence of 3 metabolic comorbidities of diabetes, hypertension, and dyslipidemia. See chapter 12 of the book

4 Acute myeloid leukemia

See chapter 11 of the book

5 Chronic granulomous disease

See chapter 13

6 rhDNase for treatment of cystic fibrosis

Original analysis in the 2000 book, updated in chapter 13.

7 Primary Biliary Cirrhosis

We will use this to explore time-dependent covariates and absolute risk. See chapter 14.

8 Mayo Clinic Study of Aging

This is the most complex data set and raises the most interesting questions. Chapter 14.

```
library(mshmm)
dim(dementia)
[1] 25671    12
summary(dementia)
```

ptnum	age	male
Min. : 1	Min. : 49.80	Min. : 0.0000
1st Qu.: 1091	1st Qu.: 71.80	1st Qu.: 0.0000
Median : 2171	Median : 79.50	Median : 1.0000
Mean : 2168	Mean : 78.15	Mean : 0.5077
3rd Qu.: 3252	3rd Qu.: 85.80	3rd Qu.: 1.0000
Max. : 4317	Max. : 104.10	Max. : 1.0000

educ	any.e4	cmc
Min. : 0.00	Min. :0.0000	Min. :0.000
1st Qu.:12.00	1st Qu.:0.0000	1st Qu.:1.000
Median :14.00	Median :0.0000	Median :2.000
Mean :14.37	Mean :0.2654	Mean :2.385
3rd Qu.:16.00	3rd Qu.:1.0000	3rd Qu.:3.000
Max. :20.00	Max. :1.0000	Max. :7.000
NAs :9		NAs :5623

dx	centiloid	p.tau181
alive : 2680	Min. : -17.00	Min. : 3.164
CU :17264	1st Qu.: 9.00	1st Qu.: 9.880
MCI : 2357	Median : 16.00	Median :13.380
dementia: 395	Mean : 31.14	Mean :15.138
nondem : 1074	3rd Qu.: 38.00	3rd Qu.:18.006
off-dem : 430	Max. :213.00	Max. :82.527
death : 1471	NAs :22716	NAs :24671

hpv1	memory	global
Min. : 3.706	Min. : -3.25859	Min. : -6.27658
1st Qu.: 6.744	1st Qu.: -0.95351	1st Qu.: -0.97472
Median : 7.285	Median : -0.01048	Median : -0.08626
Mean : 7.283	Mean : -0.12288	Mean : -0.24987
3rd Qu.: 7.844	3rd Qu.: 0.80040	3rd Qu.: 0.66211
Max. :10.199	Max. : 3.01614	Max. : 3.44065
NAs :20307	NAs :6759	NAs :8201

9 Data

The survival package accepts two kinds of input data: counting process and timeline style. The mshmm and msm packages require timeline style.

```
pdat0 <- subset(pbcseq, !duplicated(id))
pdat0$age10 <- pdat0$age/10
pdat0$bili3 <- cut(pdat0$bili, c(0,1,5,100), c("normal", "1-4", ">4"))
pdat0$death <- 1*(pdat0$status ==2)

# Create the time1, time2 dataset
pdata <- tmerge(pdat0[,c(1,4:6)], pdat0, id=id,
               death = event(futime, death),
```

```

options= list(tstart="day1", tstop="day2"))

# I have arbitrarily decided not to make use of any new lab within 7
# days of death. This loses 33 bilirubin values.
ptemp <- subset(pbcseq, !(status ==2 & (fuptime <= (day + 7))))
ptemp$bili3 <- cut(ptemp$bili, c(0, 1, 4, 100), c("normal", "1-4", ">4"))
ptemp$agrp <- cut(-ptemp$albumin, c(-100, -3.5, -3, 0),
                  c("normal", "3-3.5", "<3"))
pdata <- tmerge(pdata, ptemp, id=id, edema = tdc(day, edema),
               bili= tdc(day, bili), albumin = tdc(day, albumin),
               protime = tdc(day, protime),
               bili3 = tdc(day, bili3), b3= event(day, as.numeric(bili3)),
               agrp = tdc(day, agrp))
subset(pdata, id<3, c(id, trt, sex, edema, bili, day1, day2, death))

```

	id	trt	sex	edema	bili	day1	day2	death
1	1	1	f	1.0	14.5	0	192	0
2	1	1	f	1.0	21.3	192	400	1
3	2	1	f	0.0	1.1	0	182	0
4	2	1	f	0.0	0.8	182	365	0
5	2	1	f	0.0	1.0	365	768	0
6	2	1	f	0.0	1.9	768	1790	0
7	2	1	f	0.5	2.6	1790	2151	0
8	2	1	f	1.0	3.6	2151	2515	0
9	2	1	f	1.0	4.2	2515	2882	0
10	2	1	f	1.0	3.6	2882	3226	0
11	2	1	f	1.0	4.6	3226	5169	0

- For this subject: id
- Over this interval: (day1, day2]
- These are the covariates: trt, sex, edema, bili
- The interval ends with alive/dead transition: death

```

# Now create the multistate dataset using bilirubin groups
# 1. By definition, bilirubin can only change on 'pbcseq' days
# 2. Don't count two visits in the same state as a transition
temp <- with(pdata, ifelse(death, 4,

```

```

        ifelse(as.numeric(bili3)==b3, 0, b3)))
pdata$bstate <- factor(temp, 0:4, c("none", "normal", "1-4", ">4", "death"))
pdata$year1 <- pdata$day1/365.25
pdata$year2 <- pdata$day2/365.25

```

```

survcheck(Surv(year1, year2, bstate) ~ 1, pdata, id=id, istate=bili3)

```

Call:

```

survcheck(formula = Surv(year1, year2, bstate) ~ 1, data = pdata,
          id = id, istate = bili3)

```

Unique identifiers	Observations	Transitions
312	1930	435

Transitions table:

	to				
from	normal	1-4	>4	death	(censored)
normal	0	90	3	10	77
1-4	63	0	107	22	60
>4	1	31	0	108	35
death	0	0	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count						
state	0	1	2	3	4	5	6
normal	259	42	11	0	0	0	0
1-4	215	76	18	3	0	0	0
>4	215	86	9	2	0	0	0
death	172	140	0	0	0	0	0
(any)	91	98	70	30	13	5	5

Timeline data

- Subject id and time
- Covariates and/or status observed at that time
- Potentially lots of missing values
- Simple survival: 2 rows per subject

```

pdata2 <- subset(pbcseq, select= -c(hepato, spiders, chol, alk.phos, ast,
                                   platelet, futime, status))

pdata2$death <- 0
pdata2$bili3 <- cut(pdata2$bili, c(0, 1, 4, 100), levels(pdata2$bili3))

temp <- subset(pbcseq, !duplicated(id), c(id, futime, status))
temp$day <- temp$futime
temp$death <- 1*(temp$status==2)

pdata2 <- merge(pdata2, temp, all=TRUE)
subset(pdata2, id < 3, c(id, day, death, trt, age, sex, protime))
  id day death trt age sex protime
1  1   0     0  1 58.76523 f   12.2
2  1 192     0  1 58.76523 f   11.2
3  1 400     1 NA      NA <NA>   NA
4  2   0     0  1 56.44627 f   10.6
5  2 182     0  1 56.44627 f   11.0
6  2 365     0  1 56.44627 f   11.6
7  2 768     0  1 56.44627 f   10.6
8  2 1790    0  1 56.44627 f   11.3
9  2 2151    0  1 56.44627 f   11.5
10 2 2515    0  1 56.44627 f   11.5
11 2 2882    0  1 56.44627 f   11.5
12 2 3226    0  1 56.44627 f   11.5
13 2 5169    0 NA      NA <NA>   NA
temp <- with(pdata2, ifelse(!duplicated(id, fromLast=TRUE), 4*death,
                           as.numeric(bili3)))
pdata2$bstate <- factor(temp, 0:4,
                        c("none", "normal", "1-4", ">4", "death"))
survcheck(Surv2(day, bstate) ~ 1, pdata2, id=id)

Error in 'survcheck2()':
! REAL() can only be applied to a 'numeric', not a 'integer'

survcheck(Surv2(as.numeric(day), bstate) ~ 1, pdata2, id=id)
Call:
survcheck(formula = Surv2(as.numeric(day), bstate) ~ 1, data = pdata2,
          id = id)

```

Unique identifiers	Observations	Transitions
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312

1945

438

Transitions table:

from	to				
	normal	1-4	>4	death	(censored)
normal	0	91	3	9	77
1-4	63	0	109	21	60
>4	1	31	0	110	35
death	0	0	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count						
state	0	1	2	3	4	5	6
normal	259	42	11	0	0	0	0
1-4	214	77	18	3	0	0	0
>4	213	88	9	2	0	0	0
death	172	140	0	0	0	0	0
(any)	91	96	72	29	14	5	5

Timeline data sets are easier to create, and I suspect that timeline form will become ascendant.

What about missing values? Say that someone has a row, in the middle of follow-up, with a missing bilirubin. If that row is removed survcheck will fail, and it should.

What if we have a 3 state model, and the covariate e4pos is part of the model for ND:dementia but not for ND:death or dementia:death. Should subjects who are missing e4 be used for part of the model and not others?